

The Operational Consequences of Private Equity Buyouts: Evidence from the Restaurant Industry

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How do private equity firms affect their portfolio companies? We document operational changes in restaurant chain buyouts using comprehensive health inspection records. Store-level operational practices improve after private equity buyout, as restaurants become cleaner, safer, and better maintained. Supporting a causal interpretation, this effect is stronger in chain-owned stores than in franchised locations—"twin" restaurants over which private equity owners have limited control. These changes are particularly apparent when private equity partners have prior industry experience. The results suggest that by bringing in industry expertise, private equity firms improve firm operations. (*JEL* G24, G32)

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The private equity industry has grown tremendously over the last thirty years, reaching \$1.6 trillion in global transaction value between the years 2005 and 2007 (Kaplan and Stromberg 2008). While the accumulated evidence concludes that private equity (PE) returns outperform public stock markets (e.g., Acharya et al. 2013; Harris, Jenkinson, and Kaplan 2014), how private equity firms achieve these returns is less clear. Do PE investors actively affect portfolio firm activities and engage in operational engineering? Early evidence from the 1980s private equity wave suggests this is the case and finds a correlation between PE ownership and improved accounting performance (e.g., Kaplan 1989a; Smith 1990). However, evidence from more recent activity documents,

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at best, only modest increases in firm operating performance (Guo, Hotchkiss, and Song 2011; Acharya et al. 2013; Kaplan and Stromberg 2008). In this paper we offer a new approach to explore the operational consequences of PE firms through an in-depth analysis in the restaurant industry.

Private equity returns can be achieved without any direct or active involvement in portfolio companies. Financial engineering may be one strategy. Kaplan (1989b), for example, finds that reduced taxes from higher leverage can generate a significant portion of firm value. Similarly, Axelson et al. (2013) find that credit market conditions strongly influence private equity activity. PE firms might also be simply skilled at identifying undervalued companies or those that are about to experience a positive shift in trajectory. Guo, Hotchkiss, and Song (2011) and Acharya et al. (2013) provide evidence consistent with this hypothesis. Both find that in U.S. public-to-private transactions, targets experience only modest increases in operating performance but generate large financial returns to private equity investors. This suggests that private equity firms are able to buy low and sell high.¹ Hence, our null hypothesis is that PE firms do not actively affect firm operations.

Even if operational engineering is part of the private equity toolkit, whether this activity impacts portfolio companies positively or negatively is unclear. Critics often accuse PE firms of practicing “strip and flip” strategies, reducing long-term investment to increase immediate cash flow. A focus on short-term financial goals may adversely affect customers, employees, and firm viability in the future. Alternatively, PE firms’ active involvement may improve firm operations if they bring in industry expertise and knowledge and thus add value to portfolio companies.

We find that PE firms take an active and positive operational role in their portfolio companies. The operational changes we observe require monitoring, training, and better alignment of worker incentives, suggesting PE firms improve management practices throughout the organization. We conduct an analysis in the restaurant industry, where unique institutional features and detailed store-level data allow us to tackle two critical obstacles to drawing such conclusions.

First, it is necessary to identify whether postbuyout changes are indeed caused by PE involvement. Otherwise, any apparent operational engineering simply may be associated with the PE selection process. In the private equity literature, absent random variation in private equity buyouts, the standard approach compares the performance of PE-backed firms with that of observably similar untreated firms. This may lead to biased estimates, however, due to differences unobservable to the researcher, such as better future prospects. Our approach to address this selection concern exploits the franchise (dual ownership) structure pervasive in the restaurant industry in which, within

¹ Similarly, Bargeron et al. (2008) find that private equity firms pay lower premiums in cash acquisitions than do public company buyers.

a single chain, stores can be owned by either the parent company or a franchisee. While stores directly owned by the chain are fully vertically integrated, franchisees are legally independent entities contractually bound to keep features, such as brand, menus, and appearance, the same as in stores directly owned by the parent. Customers are unable to distinguish between company-owned and franchised units, and thus both store types likely experience similar trends and unobserved shocks (Mathewson and Winter 1985). Headquarters, however, have *limited* ability to influence franchisee decision making beyond what the franchisee contracts specify (Kidwell, Nygaard, and Silkoset 2007; Vroom and Gimeno 2007). Hence, this setting allows us to compare essentially twin stores that differ in their ownership structure and thus degree of PE influence at the time of the buyout. If PE firms actively affect the operations of their portfolio companies, these changes will manifest more strongly in the directly owned stores, especially when parent and franchisee interests are not congruent.

Operational data availability presents a second obstacle. Prior literature has primarily focused on the financial statements of PE targets that either went public or issued public debt, but such an approach may lead to biased estimates.² Even absent these biases, financial statements only shed light on aggregate firm performance, which may be too crude a measure to capture the mechanisms driving changes in firm operations. In this paper we exploit store-level data from health inspection records. Because of concerns for public health, all restaurants in the United States are regulated and subject to periodic surprise inspections, as guided by the U.S. Food and Drug Administration (FDA).³ Health inspections provide a detailed view of operational practices and routines employed by restaurants. Such practices relate to food handling, kitchen and nonkitchen maintenance, employee training and consumer advising. Examples include ensuring that food is held at the proper temperature, adequate personal hygiene of workers, toxic substances are properly stored, and food surfaces are sanitized. We compile every restaurant inspection conducted in Florida between 2002 and 2012.⁴ Private equity firms acquired 118 restaurant chains with a presence in Florida for this time period, accounting for approximately 3,400 individual restaurants out of close to 50,000 in operation.

We first employ a difference-in-differences analysis to explore the overall treatment effect of private equity firms on operational practices. Our key finding is that restaurants commit fewer health violations after being acquired by

² Exceptions are papers that focus on nonfinancial performance margins, such as innovation (Lerner, Sorensen, and Strömberg 2011), employment (Davis et al. 2014), and tax return data (Cohn and Towery 2013)

³ Each year in the United States roughly 48 million people get sick, 128,000 are hospitalized, and 3,000 die of foodborne diseases. Most of these outbreaks originate from commercial food facilities (Collins 1997).

⁴ Health inspections in the United States are commonly conducted at the county level. Each county has its own inspection standards and grading system, making cross-county health inspection comparisons difficult. We conduct the study in Florida because Florida health inspections are conducted at the state level, allowing for a consistent comparison of inspection outcomes across a larger sample.

a private equity firm. Improvements are concentrated in those practices whose potential hazards are deemed by the FDA most dangerous for customers. The effect remains strong with store-level fixed effects and when we control for changes in the number of employees and number of seats per store. The use of store-level data allows us to include ZIP-code-by-year fixed effects to control for time-varying customer demographics and local demand shocks. In addition, we show that pre-existing trends in health-related violations do not exist before private equity takes over and that the treatment effect steadily increases in the five years after the private equity buyout.

These operational practices matter for the firm and its customers. Jin and Leslie (2003) show that a reduction in violations, triggered by the introduction of hygiene quality grade cards in Los Angeles, is associated with improved store revenue. Moreover, they find that health inspection outcomes are correlated with public health and foodborne illness hospitalizations. In our sample, we find that health inspection outcomes are significantly correlated with chain reputation, measured by customer reviews posted on Yelp.com. This is consistent with Worsfold (2006), who highlights the importance of food-hazard safety and sanitation as key drivers of customers' restaurant choice. We also find that deterioration in such operational practices is correlated with future restaurant closures, a proxy for poor store profitability.

To further explore whether PE firms actively engage in operational engineering, we exploit the dual ownership structure in the restaurant industry. We find that improvements in operational practices are concentrated in directly owned restaurants in which private equity firms are more influential. We further show that both directly owned stores and franchisees experience similar pre-existing trends prior to the PE buyout. These results indicate the active involvement of private equity firms in the operations of their portfolio companies.

How do PE firms achieve these operational improvements? One possibility is increasing the number of workers employed within a store. However, we find a slight decline in a store's employee count postbuyout, consistent with Davis et al. (2014). We also find that PE firm human capital impacts the degree of improvement. Specifically, buyouts in which the lead PE firm partners have prior restaurant industry experience show the greatest reduction in health violations, and thus PE firms appear to deliver operational know-how. These findings are consistent with the survey evidence of Bloom, Sadun, and Van Reenen (2009), who find that PE-owned companies are associated with managerial best practices and with Gompers, Kaplan, and Mukharlyamov (2014), who find that PE firms claim to add operational value to their portfolio companies.

Our paper is closely related with that of Davis et al. (2014), who explore establishment-level data and find that PE-backed firms in the manufacturing sector experience productivity gains that mainly arise from an accelerated exit of less productive establishments and greater entry of more productive ones. Our paper complements theirs in several ways. First, we find that

operational improvements occur *within* an establishment. Second, we rely on institutional settings in the restaurant industry to mitigate concerns about causality. This identification strategy is important to reject the null hypothesis under which PE firms are passive investors skilled at identifying firms facing a positive trajectory. Finally, our results provide evidence of the underlying mechanism, suggesting that operational improvements occur through application of managerial experience and expertise within the firm.

More broadly, our paper contributes to an extensive literature on the effects of private equity ownership on portfolio companies' outcomes (e.g., Boucly, Sraer, and Thesmar 2011; Cohn and Towery 2013; Davis et al. 2014; Guo, Hotchkiss, and Song 2011; John, Lang, and Netter 1992; Kaplan 1989a; Lichtenberg and Siegel 1990). We deviate from the literature by focusing on detailed operational practices employed by PE investors to explain how they achieve improvements. In addition, we propose a new approach to identify the causal effect of private equity buyouts on portfolio companies.⁵ Similar to Acharya et al. (2013), we find that partner background relates to the performance of the portfolio company.

Finally, our findings are related to several papers that explore the implications of private equity on other stakeholders of the firm, such as employees (Davis et al. 2014; Agrawal and Tambe 2014), customers (Matsa 2011), and industry competitors (Chevalier 1995a, 1995b; Bernstein et al. 2014). Our paper also contributes to the theoretical and empirical literature on delegation within organizations (e.g., Aghion and Tirole 1997; Hertzberg, Liberti, and Paravisini 2011; Liberti, Seru, and Vig 2012), illustrating that various segments within the organization may be differently affected in an acquisition, depending on their degree of delegation, autonomy, and control.

1. Hypothesis Development

Private equity returns can be achieved without any direct or active involvement in portfolio company operations. They may increase target leverage, employing a financial engineering strategy to reduce taxes. PE firms may be skilled at identifying undervalued companies and time their acquisitions accordingly. We explore whether private equity firms go beyond these tactics and engage in operational engineering that directly impact their portfolio companies. Hence, we begin with the following null hypothesis:

Hypothesis 0: Private equity firms have no operational impact on their target firms.

To reject this hypothesis, we must first show, as a necessary condition, that target store-level operations indeed change after private equity takeover. This

⁵ The standard approach in the literature is to match PE-backed firms with control firms using several observable characteristics. Such counterfactuals generate unbiased estimates only under the identifying assumption that these characteristics are precisely the ones that led PE firms to invest in the portfolio company in the first place.

is not sufficient, however. Private equity acquisitions may coincide with a firm-level or industry-wide shock to restaurant operations that would have occurred regardless of PE presence. However, stores within the same chain are likely to be similarly affected by such unobserved shocks. Therefore, to strengthen the claim that the PE treatment is causal, we generate hypotheses that predict a differential effect of PE treatment *within* chains by exploiting the franchise structure of the restaurant industry. We also probe the ex ante skill sets of the PE firms to sharpen our conclusions.

Restaurant chains generate their revenue from multiple stores spread across a geographic area. These stores, while similar by all outward appearances, can have different organizational structures: company-owned or franchised. The identification strategy in this paper exploits the restaurant industry's reliance on this dual ownership model. In a franchising arrangement, the parent company (franchisor) sells a business format, typically including the brand name, menus, operating strategies, and design concepts, such as building layout and worker uniforms, to a franchisee. In return for an "off-the-shelf" business, the franchisee supplies capital for the restaurant and pays royalties to the franchisor, typically based on the sales of the franchised outlet. Franchises must maintain uniformity across units to preserve the brand and customer experience (Caves and Murphy 1976). Their appearance is similar and customers cannot easily distinguish between the two ownership types (Mathewson and Winter 1985).

A franchise is a legally independent business not vertically integrated with the parent company. Obligations to headquarters only arise through contractual agreements, which typically last twenty years and are rarely renegotiated (Lafontaine and Shaw 1999). Such contracts are hard to modify and are often loosely written to allow franchisees to adapt to local markets (Bradach 1997; Sorenson and Sørensen 2001). Because franchisees are the residual claimants of their business, they have discretion over the implementation of actions not explicitly contracted with the parent firm.

Franchising can be thought of as a form of delegated authority as in Aghion and Tirole (1997). In their model, a benefit of delegating formal authority to an agent is an increase in the agent's initiative and effort, thus mitigating agency problems. The agent will take initiative to acquire relevant information because the principal will not overrule her. Delegating formal authority to the agent, however, comes at a cost. A key downside is that the principal loses control over the choice of projects undertaken by the agent. In our setting, a store can be vertically integrated through direct ownership, giving the parent firm full legal authority. Alternatively, the parent firm can delegate authority to a franchisee. The benefit is the franchisee will expend greater effort to innovate and collect local knowledge on customers and tastes and adapt the business accordingly. The cost, however, is that the franchisor's requests and innovations may not disseminate throughout the chain, as franchisees are not obligated to implement changes outside of their contract.

Evidence for the limited control of the chain over franchisees is illustrated by the findings of Kidwell, Nygaard, and Silkoset (2007), Vroom and Gimeno (2007), and Shane (1996). Examples of the trade-off between increased effort and reduced parent firm control over delegated franchises abound for the restaurant industry. Bradach (1997) notes the experience of the CEO of Pizza Hut: "The company [direct ownership] arrangement focuses on control: a multitude of structures and systems ensures adherence to the standards and preserves uniformity. The emphasis on control, however, does not create a context hospitable to innovation and adaptation. Conversely, franchisees are bound to the chain with a relational contract, with explicit and implicit rights and obligations, which provides the motivation and autonomy to generate and experiment with new ideas. The problem is controlling the behavior of franchisees: it can be difficult to maintain uniformity and obtain franchisees' agreement to adopt system-wide adaptations." Also, "One executive said [describing company owned units], 'we essentially have a military organization . . . we tell them and they do it.'" In contrast, formal authority was not granted to the chain in the franchise arrangement. An often-repeated industry adage was "we sell franchisees and tell company managers."

In the language of Aghion and Tirole (1997), when a private equity firm takes over a chain, this is equivalent to a change in the identity of the principal, and thus, if the PE firm attempts to change operational practices, it may have less influence implementing these in franchises relative to directly owned stores.

Importantly, in addition to a PE firm having less influence over franchisees, PE firm and franchisee interests may not be fully congruent. This is because franchisees seek to maximize their own profit rather than the profit of the organization as a whole (Caves and Murphy 1976). A customer's bad experience at a franchised store may adversely affect his opinion of the entire brand, as he does not distinguish between company-owned and franchised units. The franchisee, however, does not internalize this externality—it does not matter to the franchisee if customers shun other branches. Indeed, Jin and Leslie (2009) find that franchisees free ride on the reputation of the chain by exerting less effort to maintain good hygiene.

The divergence of interests between the chain and the franchisee, combined with the lack of parent firm authority to tell franchisees what to do, suggests that within a chain, operational engineering initiatives by the private equity firm will more strongly manifest in directly owned (treated stores) than in franchised units (quasi-treated stores).

Hypothesis 1: Operational improvements initiated by the private equity owner will manifest more strongly in directly owned stores than in franchises.

Moreover, the congruence of interests between the parent firm and the franchisee may depend on the nature of the operational initiatives and should be greater when benefits primarily accrue to individual stores rather than to the

whole chain. For example, franchisees may be more likely to adopt initiatives that cut costs and improve immediate store profits than initiatives that improve quality of service, which improve chain reputation and generate rents shared by all stores.

Hypothesis 2: The gap in operational impact between directly owned stores and franchises is lower for initiatives in which the benefits accrue to the individual store, rather than to the entire chain.

Implementing successful operational change requires expertise. Bloom, Sadun, and Van Reenen (2009) find that PE-backed firms are associated with better general management practices. Acharya et al. (2013) find that PE partner background relates to the performance of portfolio companies. If private equity owners are the cause of operational change, we have a final hypothesis:

Hypothesis 3: Private equity partner operational expertise (or, more specifically, restaurant industry expertise) is associated with better operational outcomes.

We explore these hypotheses in the remainder of the paper and discuss the identification strategy in Section 3.

2. Data Description

The data in this analysis are constructed from several sources combining information on PE buyouts (CapitalIQ), health inspection records and restaurant ownership in Florida (Florida Department of Business and Professional Regulation), store-level employment (InfoUSA), and restaurant consumer reviews (Yelp.com). In this section, we describe these data and illustrate how health inspection outcomes vary across different types of restaurants and locations. We also show how inspection results correlate with customer satisfaction and restaurant closures.

2.1 Health inspections

This paper focuses on restaurant operational practices as captured by surprise and mandatory health inspections. Because of concerns for public health safety, all restaurants in the United States are subject to periodic health inspections conducted by trained specialists in food service evaluation certified by the Food and Drug Administration (FDA). Failed inspections can result in fines, suspensions, and, in extreme cases, restaurant closure.⁶ Inspection outcomes at restaurants in Florida are not easily observable by customers (unlike the hygiene-quality letter grade cards in cities like Los Angeles and New York).

⁶ These fines, however, are fairly marginal. In 2012, approximately 2% of Florida inspections resulted in fines, with a median payment of \$400.

We gather health inspection data from the Florida Department of Business and Professional Regulation encompassing every restaurant inspection conducted in the state of Florida from 2002 through 2012. U.S. health inspections are typically organized and conducted at the county level, and each county is free to use its own criteria and scoring methodology.⁷ This makes combining inspection data from different geographic locations troublesome. We use data from Florida because inspections there are conducted at the state level, thus providing the largest possible U.S. sample of inspection results using common criteria. Historical Florida records are available to 2002. Each record gives the name of the restaurant, the address, the date of the inspection, and the incidence of fifty-eight types of operational violations.

The Florida Department of Health divides violations into critical and noncritical. Critical violations are those “likely to directly contribute to food contamination, illness, or environmental degradation.” Examples include improper disposal of waste, improper temperatures for cooked or stored food, dirty restrooms, and contaminated food surfaces. Noncritical violations “do not directly relate to foodborne illness risk, but preventive measures are required.” Examples include dirty nonfood contact surfaces, inadequate lighting, and lack of clean clothes and hair restraints. A complete list of inspection violations is provided in the Appendix. Inspections primarily fall into three categories: surprise, follow-up, and initial setup. We consider only surprise inspections for this study. Follow-ups are arranged in response to violations that need to be fixed and, like initial setup inspections, occur on known dates that allow restaurants to put their best foot forward.

2.2 Other data sources

We supplement the inspection data with restaurant ownership data also from the Florida Department of Business and Professional Regulation. Restaurants need to renew licensing agreements with the state each year. These licenses, available from 2002 to 2012, provide the name and address of the owner at each restaurant. This allows us to separate restaurant branches into those directly owned by the parent brand, for which the owner name and address coincide with those of the parent firm, and those that have been franchised to independent owners. We incorporate data from InfoUSA, which makes phone calls to establishments to gather, among other data items, the number of full-time equivalent employees. This data are gathered annually. Employee count is matched to the inspection database by name, address, and geocode coordinates.⁸ We collect median county income from the Bureau of Economic and Business Research (BEBR) at the University of Florida. We also collect more than 300,000 consumer restaurant reviews, downloaded from Yelp.com.

⁷ Letter grades, numerical scores of varying ranges, and pass/fail scores are among the health inspection grading systems used across the United States.

⁸ For a detailed discussion of InfoUSA and an illustration of its wide coverage, see Kennel and Li (2009).

To determine which of these restaurants was acquired by private equity firms, we download all Leveraged Buyout, Management Buyout, and Secondary Buyouts in the restaurant industry from Capital IQ. We research each deal to find the names of the restaurant chains involved and record the date the deal closes. We find 118 PE transactions with 3,342 individual restaurant stores in Florida between 2002 and 2012.⁹

Table 1 provides summary statistics on inspections and the restaurant sample. Panel A shows that more than 21,000 eating establishments are inspected roughly two times per year.¹⁰ The mean number of critical and noncritical violations found is 4.49 and 3.35, respectively. Panel B shows the distribution of PE transactions over time in our sample. As expected, the number of private equity deals is higher during the buyout boom period, between 2005 and 2007 (roughly 45% of deals in our sample occurred at this time). Moreover, there were few transactions during the crisis, and the number of buyouts picked up gradually thereafter. These patterns follow the overall trends in private equity activity during these years.¹¹

Panel C of Table 1 shows that chains acquired by PE firms have, on average, 175 outlets in Florida. Throughout the sample years, these stores generate 3.78 critical violations per inspection, similar to violation counts in untreated chains with at least five stores in Florida. During the sample years, treated and untreated stores of chains with at least five stores also appear similar on employee counts, seats, and local county income. At the year of the buyout, as illustrated in Column (2) of panel C, target stores appear to have a slightly higher number of violations and critical violations relative to large never-treated chains. These differences are statistically significant and motivate the identification strategy discussed below. Panel D shows that private equity is present in all types of cuisines, with a greater relative presence in hamburger chains. Panel E shows that treated and untreated chains have a similar price distribution, with about half the stores in the lowest price segment. In panel F we compare franchisees and directly owned stores, who seem to have slightly fewer violations. This is consistent with findings reported by Jin and Leslie (2009), who explore health inspections in Los Angeles. In addition, directly owned stores seem to be somewhat larger in terms of seats and employees per store. Interestingly, both directly owned stores and franchisees seem to be located in counties with similar income.

⁹ Out of all the PE targets, only one chain exclusively operates in Florida. All other chains have operations in multiple states.

¹⁰ We include only those stores for which we have employee and seat counts. The total number of inspections in 2002 is small because the data do not cover the entire year.

¹¹ In Table A1 in the Appendix we provide summary statistics on the type of the PE transaction. We divide the deals into four categories based on the taxonomy of Kaplan and Stromberg (2008). More than 40% of the deals in our sample are private-to-private deals, while 22% of the deals are public-to-private, and 20% are secondary buyouts. The distribution of types of PE deals is similar to that of Davis et al. (2014).

Table 1
Inspection summary statistics*A. Health inspections over time*

Year	Inspections conducted	Restaurants inspected	Inspections per restaurant	Average critical violations	Average noncritical violations
2002	20,308	15,656	1.30	1.67	2.30
2003	42,959	19,051	2.25	2.17	3.04
2004	35,087	18,901	1.86	2.31	3.60
2005	36,052	19,335	1.86	2.80	4.13
2006	40,224	20,702	1.94	4.85	4.61
2007	40,748	21,026	1.94	6.62	3.93
2008	47,350	21,771	2.17	6.19	3.88
2009	57,537	24,002	2.40	5.34	3.20
2010	63,388	27,228	2.33	5.25	3.19
2011	65,971	27,413	2.41	4.70	2.72
2012	60,329	25,783	2.34	4.56	2.69
Mean	46,359	21,897	2.12	4.49	3.35

B. Private equity deals over time

Year	PE transactions	Stores acquired
2002	5	4.24%
2003	5	4.24%
2004	9	7.63%
2005	10	8.47%
2006	21	17.80%
2007	22	18.64%
2008	8	6.78%
2009	2	1.69%
2010	17	14.41%
2011	10	8.47%
2012	9	7.63%
Total	118	100.00%

C. Restaurant characteristics

	Ever treated stores		Treated stores at buyout year		Never treated stores		Never treated stores (at least five stores)	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Violations	7.20	5.29	7.75	5.57	8.19	5.25	6.91	5.25
Critical violations	3.78	3.21	4.25	3.46	4.69	3.31	3.80	3.31
Noncritical violations	3.42	2.89	3.50	2.86	3.50	2.71	3.11	2.71
Log seats	4.16	1.28	4.16	1.24	3.77	1.69	3.56	1.69
Log employees	3.07	0.84	3.01	0.87	2.39	1.03	2.83	1.03
Chain units (in Florida)	175.27	168.18	192.80	201.07	109.80	331.84	282.08	331.84
Log county income	10.50	0.20	10.54	0.19	10.53	0.20	10.51	0.20

D. Restaurant cuisines (%)

Cuisine	Treated stores at buyout year	Never treated stores	Never treated stores (at least five stores)	All stores
American	18.21	26.84	19.89	25.49
Asian	0.92	10.77	9.95	9.23
Chicken	12.47	6.54	9.01	7.39
Donut, ice cream, beverage	5.01	5.01	5.61	5.02
Hamburgers	33.77	11.37	16.02	14.80
Other ethnic	6.34	8.55	6.61	8.28
Pizza, pasta, and Italian	7.62	13.47	13.51	12.56
Sandwiches, soups, and salads	8.50	12.96	15.82	12.31
Steak, seafood, and fish	7.14	4.49	3.57	4.91

(continued)

Table 1
Continued

E. Restaurant price categories (%)

Average restaurant check	Ever treated stores	Never treated stores	Never treated stores (at least five stores)	All stores
> \$7	51.58	32.15	43.11	35.19
\$7 to \$10	29.43	26.43	27.67	26.90
\$10 to \$15	14.92	36.00	26.24	32.70
< \$15	4.08	5.41	2.98	5.20

F. Comparison of franchisees and directly owned stores

	Directly owned		Franchisees		All Stores	
	Mean	SD	Mean	SD	Mean	SD
Violations	5.85	4.37	5.98	4.97	5.94	4.77
Critical violations	2.93	2.72	3.14	3.17	3.07	3.02
Noncritical violations	2.89	2.37	2.80	2.49	2.83	2.45
Log seats	3.84	1.70	3.43	1.76	3.57	1.75
Log employees	2.85	1.23	2.67	1.14	2.73	1.18
Log county income	10.41	0.21	10.41	0.22	10.41	0.22

This table summarizes the Florida restaurant health inspection data. According to the Florida Department of Health, critical violations are those “likely to directly contribute to food contamination, illness, or environmental degradation.” Noncritical violations “do not directly relate to foodborne illness risk, but preventive measures are required.” Only surprise inspections are counted. Panel A shows the average level and frequency of all inspections conducted in Florida in the respective years. In panel B, *PE transactions* counts the number of chains acquired; *Stores acquired* counts the individual outlets in Florida associated with those chains. Panel C summarizes violation, store, and environment characteristics for all store-years in the sample. *Ever treated stores* are individual stores that have a private equity parent at some point in the sample. Stores with *At least five stores* are part of a chain with at least five units in Florida. *Seats* and *Employees* count the number of seats and full-time equivalent employees in the store-year. *Chain units* is the number of total stores in Florida with the same brand. County income is the median income in the store’s county that year. Panels D and E show the distribution of cuisines and average check sizes per diner across the treated and untreated stores. Panel F summarizes the characteristics of store-years under direct parent and franchise ownership.

2.3 Correlation between health inspections and other restaurant characteristics

To better understand the role of operational practices, we study in this section the relationship between health violations and other important restaurant characteristics. We start by exploring the general determinants of health violations in restaurants. In Table 2, Column (1), the dependent variable is critical violations and we include year fixed effects. We find that larger stores—those with more seats and employees—have more violations. Larger stores may be more difficult to manage and have more sources for violations. Richer neighborhoods see fewer violations. The more units in the restaurant chain, the better the inspection outcome. This may be evidence of professional management; a firm running multiple stores has more experience and better controls and procedures in place to monitor operational practices than does a proprietor opening a single store. By cuisine type, Asian establishments fare the worst, while donut shops, ice cream parlors, and beverage stores are the cleanest. These latter categories offer simpler items and less variety, which may explain the number of fewer violations. Column (2) adds restaurant chain fixed effects and drops chain-invariant variables. The remaining results are

Table 2
Drivers of restaurant health and cleanliness

Critical violations	Critical violations	Noncritical violations	Noncritical violations	
	(1)	(2)	(3)	(4)
Log(units in chain)	−0.238*** (0.026)		−0.112*** (0.020)	
Log(seats)	0.262*** (0.024)	0.195*** (0.027)	0.223*** (0.022)	0.141*** (0.022)
Log(employees)	0.065** (0.033)	0.079*** (0.020)	0.179*** (0.024)	0.096*** (0.015)
Log(median county income)	−0.535*** (0.101)	−0.334** (0.132)	−0.509*** (0.074)	−0.345*** (0.088)
Average check < \$7	−0.632** (0.275)		−0.232 (0.219)	
\$7 to \$10	−0.394** (0.164)		−0.095 (0.128)	
\$10 to \$20	0.220* (0.132)		0.207* (0.120)	
Cuisine type				
American: Omitted category				
Asian	1.628***		1.050***	
Chicken	0.032		0.543***	
Donut, ice cream, beverage	−0.542**		−0.530**	
Hamburgers	−0.240		−0.433	
Other ethnic	−0.101		−0.224	
Pizza, pasta, Italian	0.178		−0.136	
Sandwiches, soup, deli	−0.417*		−0.594***	
Steak, seafood	−0.263		−0.038	
Year fixed effects	X	X	X	X
Chain fixed effects		X		X
Observations	345,489	345,489	345,489	345,489
R ²	0.20	0.32	0.10	0.21

This table presents results from ordinary least-squares (OLS) regressions of general store characteristics on restaurant health inspection outcomes. An observation is a store-year. Violations are as defined in Table 1. *Units in chain* counts the total number of separate stores of that particular restaurant chain in Florida each year. *Median county income* is the median income each year in the restaurant's county. *Seats* and *Employees* are measured at the store level. *Check sizes* are average spending per diner. Standard errors are omitted for cuisine types for brevity. Standard errors are clustered by restaurant chain. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

unchanged. Higher median county income leads to fewer violations even within the same chain. Columns (3) and (4) switch to noncritical violations, and the results are nearly the same.¹²

Chain reputation may provide an important motivation for maintaining high levels of store maintenance and hygiene, as argued by Jin and Leslie (2009). To further explore this, we extract data from Yelp.com, a consumer review Web site, to explore whether health-related operational practices are correlated with overall customer satisfaction. People who register as users with Yelp by providing a valid E-mail address can leave comments on restaurants and other businesses and star ratings ranging from one to five. In Florida, Yelp reviews

¹² The number of observations in Table 2 is smaller than in later tables because cuisine type and check size are not fully populated. These characteristics are not used in the main analysis, since they are fixed and therefore drop out with restaurant fixed effects.

Table 3
Health violations customer satisfaction, and store closure

<i>A. Dependent variable = Avg Yelp stars</i>						
	(1a)	(2a)	(3a)	(4a)	(5a)	(6a)
Avg critical violations	−0.023*** (0.003)	−0.022*** (0.004)	−0.022*** (0.004)	−0.016*** (0.004)	−0.017*** (0.004)	−0.016*** (0.004)
Avg noncritical violations				−0.018** (0.007)	−0.012* (0.007)	−0.015** (0.007)
Price × Cuisine fixed Effects		X	X		X	X
Five or more reviews			X			X
Observations	3,775	3,775	1,908	3,775	3,775	1,908
R ²	0.010	0.060	0.119	0.011	0.061	0.121
<i>B. Dependent variable =Store closure</i>						
	(1b)	(2b)	(3b)			
Annual critical violations	0.00068** (0.00026)		0.00088*** (0.00029)			
Lagged annual critical violations	0.0023*** (0.00025)		0.00204*** (0.00027)			
Annual noncritical violations		−0.00030 (0.00029)	−0.00066** (0.00031)			
Lagged annual noncritical violations		0.0018*** (0.00028)	0.00073** (0.00030)			
Log(seats)	0.0076* (0.0042)	0.0082* (0.0042)	0.0076* (0.0038)			
Log(employees)	0.0098*** (0.0014)	0.0098*** (0.0014)	0.0098*** (0.0014)			
Year fixed effects	X	X	X			
Store fixed effects	X	X	X			
Observations	203,006	203,006	203,006			
R ²	0.54	0.54	0.54			

This table presents results from OLS regressions of customer satisfaction and restaurant closure on restaurant sanitation. In panel A, an observation is a restaurant chain. The dependent variable *Avg Yelp stars* is the average star rating (which can range from 1 to 5) for all reviews given to all branches in a chain in Florida in 2012 on the Web site Yelp.com. The independent variable *Avg critical violations* averages the critical violations for all inspections for all branches in a chain in 2012. The restriction *Five or more reviews* refers to the number of Yelp reviews for the chain in 2012. *Price x Cuisine fixed effects* captures the combination of cuisine type and average check size. In panel B, an observation is a store-year. The dependent indicator variable equals one if the store closed in that year. *Annual critical violations* is the average number of such violations in all (typically two) surprise inspections at that store that year. *Lagged violations* average those the year before the closure year. Standard errors are in parentheses and clustered by restaurant chain in panel B. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

are sparse before 2010 and increase significantly by 2012. We thus do not have a sufficient panel structure to examine the impact of PE on consumer satisfaction, but we exploit the cross-sectional correlation between this review-based restaurant quality measure and health violations in Table 3, panel A.

For the year 2012, we average the number of critical violations found in all inspections for all branches at the chain level. We also average the number of Yelp stars for each chain. Column (1a) shows the results of a simple univariate regression of stars on critical violations. The coefficient on critical violations is −0.023 and is highly significant. A four-violation increase (one standard deviation) is thus associated with a rating lower by 1/10 of a star. This is meaningful, given that small changes in ratings are associated with significant changes in revenue (Luca 2011). Moreover, given that few of the inspected

practices occur outside the kitchen and are thus observable to customers, the results may underestimate the importance of such operational practices to customers.

Column (2a) adds restaurant price range by cuisine fixed effects (e.g., \$10–\$15 check size, Asian). Violations and customer satisfaction are strongly negatively related, even among similar restaurants. Column (3a) shows the results of a robustness check requiring at least five Yelp reviews for a chain, and the results remain the same. Columns (4a) through (6a) add noncritical violations. These are also negatively related to Yelp scores, but the statistical significance is somewhat weaker.

The relationship between operational practices and perceived quality could be a direct effect, as customers down rate stores with poor hygiene. The correlation may also reflect that a restaurant that sustains good practices also performs better on quality dimensions, such as service and food. Both explanations suggest that changes to health inspection results may have broader implications for customer satisfaction.

Panel B of Table 3 shows that poor practices are correlated with a more direct outcome: restaurant closure, a proxy for poor financial performance.¹³ For each restaurant, we average all inspection scores received each year. We then create the dummy variable *store closure*, which equals one in the year a store closes, if it closes. Closure is defined as having no inspection record in a given year or in subsequent years. The inspection database is comprehensive, and every restaurant must be inspected at least once and usually twice each year. Thus, if no inspections occur in a given year, we assume it must have closed. In Column (1b), we regress store closure on the number of critical violations received in the year of closure, the year before, and year and store fixed effects. The coefficient on annual critical violations is significant at the 5% level. A one-standard-deviation increase in critical violations (4) is associated with a 0.3% increase in the likelihood of closure that year. This is not small, given that the unconditional likelihood of closure is 7% per year, implying a 4% increase in closure likelihood. In fact, the number of violations the prior year has more than three times this impact, suggesting that an increase of one standard deviation in critical violations is associated with an almost 15% increase in likelihood of store closure the following year. Noncritical violations, added in Columns (2b) and (3b), are not as strong a factor and do not affect the correlations of critical violations with store closures.¹⁴

Overall, this section illustrates the importance of operational practices. Such practices correlate with customer satisfaction and chain reputation. Moreover, these operational practices are correlated with store closures. This evidence

¹³ Mandatory closures enforced due to poor health inspections are rare. The closures discussed here are voluntary and decided by the restaurant owners.

¹⁴ In Table A2 in the Appendix we repeat the analysis of this section, focusing only on restaurant chains (i.e., brands with multiple units). The results remain unchanged.

complements Jin and Leslie (2003), who find that a reduction in violations, triggered by the introduction of hygiene quality grade cards in Los Angeles, is associated with improved store revenue and reduced incidence of foodborne illness hospitalizations.

3. Empirical Methodology

3.1 Initial specification

We examine the effect of private equity buyouts on health inspection outcomes using a difference-in-differences methodology. We estimate the following equation:

$$y_{ijt} = \eta_{ij} + \lambda_{kt} + \beta \times PostPE_{it} + \gamma' X_{ijt} + \varepsilon_{ijt}, \quad (1)$$

where i indexes the restaurant chain, j indexes specific store within a chain, t indexes time, y_{ijt} is the dependent variable of interest (e.g., number of health inspection violations), η_{ij} and λ_{kt} are store fixed effects and ZIP-code-by-year fixed effects, respectively, X_{ijt} are control variables (such as number of seats and employees per store), and ε_{ijt} is an error term. To allow for serial dependence of the error term, we cluster standard errors at the restaurant chain level rather than at the store level. The main coefficient of interest is β , where $PostPE_{it}$ is a dummy variable that equals one if restaurant chain i was acquired by a private equity firm at or before time t .

The identification strategy can be illustrated with a simple example. Consider a restaurant chain that was acquired by a private equity firm. The effect of this treatment on store-level operations can be measured by simply comparing health inspection violations before and after the buyout. This difference removes any time-invariant unobservable characteristics at the level of the store and is equivalent to including the store-level fixed effects outlined in the above specification. Time-invariant characteristics may include chain-level variables, such as cuisine type, price segment, or brand, and store-level characteristics, such as location.

Store-level operational practices also can be affected by time-varying events. For example, enforcement standards might change or media attention might increase customer awareness of hygiene standards. Therefore, we include a control group that consists of all the restaurants in Florida that have not (yet) been treated. If tougher inspection standards lead *all* restaurants to improve operations, treated restaurant improvement should not be credited to PE firm involvement. Because of the staggered nature of the private equity buyouts, restaurants remain in the control group until they are treated (which, for most restaurants, will never happen). We compare the changes in operational practices of treated restaurants with changes to the control group. This difference-in-differences estimates the effect of private equity buyout on restaurant operations.

Restaurants targeted by private equity may still differ from the control group in unobservable ways. If such factors are omitted from the specification,

then the relationship between private equity and operational practices may be spurious. We further discuss how our identification strategy can account for such concerns.

A key ingredient to a restaurant's success is its location. If a restaurant's area is booming or local demographics shift, operational practices may evolve to handle increased business. In that case, one may be concerned that PE firms are simply passive buyers who acquire chains located in areas experiencing an upward trajectory accompanied by improvement in operations. However, because of our detailed location information for all stores in Florida, we can control for such local shocks by including a full set of ZIP code fixed effects interacted with year fixed effects, thus separating the effects of the private equity buyout from contemporaneous local shocks. Our control group thus becomes untreated restaurants in the same ZIP code.

Shocks specific to the restaurant chain are an additional possibility. Since such shocks do not affect other restaurants in the same area, they cannot be accounted for by the inclusion of ZIP-code-by-year fixed effects. Moreover, since these are time-varying shocks, they cannot be eliminated with restaurant fixed effects. To mitigate such concerns, we explore the dynamics of operational practices around the private equity buyout. If private equity firms are capitalizing on a preexisting shock, then we should see a trend starting before the buyout. Hence, we run specifications using pre- and postevent year dummies and find no effect before the acquisition.

While the empirical strategy so far can address several important concerns, anticipation of future chain-level shocks may still lead to spurious correlation. An ideal experiment would compare two identical stores: one treated with PE ownership and the other without. The prevalence of the franchising model in the restaurant industry allows us to run a close variation of such an experiment.

3.2 Comparing franchisees and directly owned stores

To alleviate concerns that postbuyout operational changes, such as an anticipation of future chain prospects, are associated with the PE firm selection process, we compare directly owned stores with franchisees. Our identifying assumptions in this analysis are as follows: First, if an unobserved trend affects a given chain (such as a future demand shock), it *similarly affects* both franchisees and directly owned stores within the chain. This is arguably the case as they share similar characteristics, such as brand name, menus, and food preparation techniques, and they are basically indistinguishable by customers (Mathewson and Winter 1985). Second, PE firms (and chains) have more control and influence over directly owned stores than do franchisees, and thus, while directly owned stores are *treated* by PE firms, franchisees are only *quasi treated*. When PE firms attempt to restructure a chain, they can do so fully in directly owned stores and only partially in franchises. Both assumptions are supported by prior theory and empirical evidence as discussed in detail in Section 1. Taken together, differences in outcomes between the two ownership

types will reveal whether PE firms actively affect store level operations. This motivates the following specification:

$$y_{ijt} = \eta_{ij} + \lambda_{kt} + \beta \times PostPE_{it} + \gamma \times Direct_{ijt} + \delta \times PostPE_{it} \times Direct_{ijt} + \zeta' X_{ijt} + \varepsilon_{ijt}, \quad (2)$$

where i indexes restaurant chains, j indexes a specific store within a chain, t indexes time, and $Direct_{ijt}$ is a dummy variable equal to one if a store is directly owned by headquarters and zero if owned by a franchisee. All other variables are defined earlier. We also cluster standard errors at the level of the restaurant chain to allow error dependence between stores within the same chain. In this specification, the coefficient of interest is δ , which compares the effect of PE buyouts on y in directly owned stores relative to franchisees.

As in Equation (1), this specification includes store fixed effects, η_{ij} , which capture time-invariant heterogeneity, removing chain-level variation common to both directly owned stores and franchisees, as well as store-level variation.¹⁵ Hence, the specification allows the interpretation of δ as a comparison between directly owned stores and franchisees within the same chain. The ZIP-code-by-year fixed effects, λ_{kt} , allow us to isolate δ from time-varying unobserved local shocks, as discussed earlier.

It is worth emphasizing that we do not assume that franchisees are unaffected by PE firms. Rather, due to differences in objectives (such as that franchisees do not internalize the externalities they may impose on the chain), franchisees are more likely to resist adopting changes to operational practices suggested by the chain (or PE firm) relative to directly owned stores, as we discuss in Section 1. Finally, we do not assume that franchisees are randomly assigned within a chain. This may be a concern if factors that determine franchisee assignment also impact operational practices at the time of the buyout. We discuss this in the next section.

3.3 Determinants of the decision to franchise versus directly own

Why are certain stores company owned, and do these same underlying reasons drive operational practices following the private equity buyout? The franchising literature often relies on theoretical agency arguments, specifically moral hazard models, to derive predictions for whether restaurants or retailers are directly owned or franchised (Lafontaine and Slade 2007). One prediction from such models is that franchising is more common when individual store effort matters more. In this case, compensation in the form of residual profit may be needed to induce the necessary effort (Norton 1998). Such models also predict that company ownership is more likely when store size and initial setup costs are bigger (Lafontaine 1992) and daily operations are more complex (Yeap 2006).

¹⁵ Despite the inclusion of store fixed effects, the $Direct_{ijt}$ dummy variable remains in the estimation, as some restaurants change their ownership status over time.

These cross-sectional predictions can help explain variations in franchising tendency between chains, but given the similarity between a brand's stores, these models do not seem to explain ownership variation within a chain.

Within a chain, costly monitoring may explain the decision to own or franchise a particular location (Lafontaine and Slade 1996). Some evidence illustrates that locations more distant from headquarters are more likely to be franchises, as monitoring is more difficult (Brickley and Dark 1987; Minkler 1990). If such regions are more likely to experience, for example, an economic boom following a private equity buyout, that may be a concern. However, the inclusion of ZIP-code-by-year fixed effects mitigates such concerns. We also do not find differences in preexisting trends around the PE transaction between franchisees and directly owned stores.

A key difference between directly owned stores and franchises is that franchisees do not internalize the externalities generated by maintaining high-quality service, as the cost of maintaining quality is private, while benefits accrue to all members of the chain. Franchisees may free ride on the chain's brand name and use lower-quality inputs to reduce costs (Jin and Leslie 2009; Klein 1995). Such free-riding behavior may be exacerbated in locations with nonrepeat customers, such as highway exits. In such locations, the chain may therefore choose to operate its own stores. While supporting evidence for direct ownership at highway exits is mixed (Lafontaine and Slade 2007), concerns that hidden factors at such locations may be correlated with PE buyout occurrences remain. The inclusion of ZIP-code-by-year fixed effects mitigates this concern. Moreover, in a robustness test discussed below, we directly control for highway exit locations and find that results are unaffected.

4. Results

4.1 Private equity and restaurant operational practices

We start by exploring the relationship between private equity ownership and operational practices as a test of our null hypothesis in which PE firms have no operational impact on their portfolio companies. Panel A of Table 4 implements specification (1) discussed in Section 3.1 in which the dependent variable is the number of critical violations. Column (1a) includes chain fixed effects and year fixed effects. The coefficient on *PostPE* is -0.664 and significant at the 1% level. Since the average number of critical violations is 4.5, this is a sizable decline of 15%. In Column (2a) we add the number of restaurant seats and employees as controls, motivated by Table 2. The larger the restaurant, the more critical violations, but the *PostPE* coefficient retains similar magnitude. This suggests that the improvements in operational practices following the buyout are not driven by changes in restaurant size or employee count. In Columns (3a) and (4a), we replace chain fixed effects with individual store fixed effects that capture constant unobserved store-level heterogeneity. The coefficient on *PostPE* remains the same with slightly lower significance, now at the 5% level, in this stricter test.

Table 4
Violations under private equity ownership

A. Dependent variable =Critical violations

	(1a)	(2a)	(3a)	(4a)	(5a)	(6a)
PostPE	-0.664*** (0.240)	-0.648*** (0.240)	-0.630** (0.251)	-0.626** (0.251)	-0.614** (0.254)	-0.612** (0.253)
Log(seats)		0.212*** (0.025)		0.239*** (0.050)		0.178*** (0.053)
Log(employees)		0.067*** (0.016)		-0.028 (0.018)		-0.027 (0.019)
Chain fixed effects	X	X				
Store fixed effects			X	X	X	X
Year fixed effects	X	X	X	X		
ZIP × Year fixed effects					X	X
Observations	541,147	541,147	541,147	541,147	541,147	541,147
R ²	0.129	0.130	0.135	0.135	0.535	0.535

B. Dependent variable =Noncritical violations

	(1b)	(2b)	(3b)	(4b)	(5b)	(6b)
PostPE	0.056 (0.140)	0.069 (0.141)	0.085 (0.157)	0.084 (0.156)	0.022 (0.150)	0.023 (0.150)
Log(seats)		0.155*** (0.021)		0.006 (0.0415)		0.037 (0.036)
Log(employees)		0.084*** (0.013)		-0.007 (0.012)		-0.002 (0.011)
Chain fixed effects	X	X				
Store fixed effects			X	X	X	X
Year fixed effects	X	X	X	X		
ZIP × year fixed effects					X	X
Observations	541,147	541,147	541,147	541,147	541,147	541,147
R ²	0.034	0.036	0.034	0.034	0.470	0.470

This table presents results from OLS regressions of restaurant inspection results on private equity ownership and store characteristics. An observation is an inspection on a specific date at a specific store. The dependent variables are defined in Table 1. *PostPE* is a dummy variable that equals one if the inspection occurs after the restaurant has been acquired by a private equity firm. *Log(Seats)* and *Log(Employees)* count the number of seats and full-time equivalent employees at the restaurant in the year of the inspection. *ZIP × Year fixed effects* use the ZIP code of each restaurant. Standard errors are in parentheses and clustered by restaurant chain. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

We introduce an even more precise counterfactual in Columns (5a) and (6a) by replacing year fixed effects with ZIP-code-by-year fixed effects.¹⁶ This specification compares PE-treated restaurants to competitors in the same ZIP code. Therefore, restaurants are serving the same time-varying demographics and are subject to similar local demand shocks. Even after adjusting for location, critical violations still decline after PE entry, and the effect is statistically significant. In fact, in Table A3 in the Appendix, we find that these improvements are concentrated in practices related to food handling, as well as training and consumer advising. This emphasizes the skill channel through which PE firms

¹⁶ It is computationally difficult to estimate a regression that has so many layers of fixed effects. Fortunately, algorithms have been developed to handle such high-dimensional fixed effect regressions. Following Gormley and Matsa (2014), we use the iterative algorithm of Guimaraes and Portugal (2010).

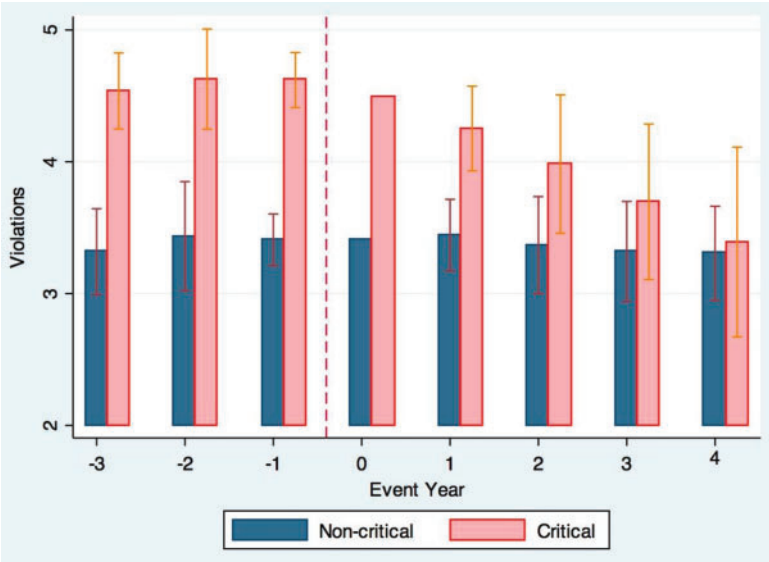


Figure 1
Critical and noncritical violations around private equity deal date

This figure plots the coefficients and 95% confidence interval bands of regressions of critical and noncritical violations on event-year dummy variables around the date private equity acquires a restaurant. Additional control variables are restaurant fixed effects, ZIP-code-by-year fixed effects, number of employees, and number of seats. Standard errors are clustered at the level of the chain. Event-year zero is the omitted variable, corresponding to inspections that occur from 1 to 365 days after the deal close date. The average number of critical and noncritical violations was added to coefficients in the graph.

operate, as changes in food handling practices, for example, are unlikely to be achieved simply by capital reallocation. We discuss this further in Section 5.3.

Panel B of Table 4 replaces critical with noncritical violations. In all six specifications, the effect of private equity management is essentially zero. Noncritical violations have a much smaller effect on health outcomes and, as illustrated in Table 3, have a weaker effect on customer satisfaction and store closures. Hence, improvements appear to be concentrated where operational practices matter more.

Figure 1 shows the path of critical and noncritical violations around private equity takeover. The light bars plot the coefficients of a regression in which critical violations are regressed on private equity entry event year dummies.¹⁷ Violations are flat in the three years before PE entry. Thus, there does not appear to be a predeal trend. This helps mitigate endogeneity concerns that private equity firms were simply capitalizing on a trend of improved operational practices. The decline in critical violations then steadily occurs over the

¹⁷ This regression includes store fixed effects, ZIP-code-by-year fixed effects, log number of seats, and log number of employees. The regression results are in the Appendix, Table A4. The average number of critical violations was added to coefficients in the graph to illustrate the relative size of the coefficients.

subsequent four years, becoming statistically significant in year three onward. This is consistent with anecdotal evidence on the gradual speed of operational change in restaurants (Gompers, Mugford, and Kim 2012). The dark bars plot the evolution of noncritical violations. The pattern appears flat before and after the PE buyout.¹⁸ The evidence thus far is consistent with a rejection of our null hypothesis that PE firms do not have operational impact.

4.2 Operational practices: Twin restaurants analysis

In this section, we turn to the “twin restaurant” analysis, comparing franchises and directly owned stores in a test of hypotheses 1 that postulates that operational changes are likely to be more apparent within a chain in which the PE firm has more control and more influence. For this test, we are interested in chains that have a mixture of both franchises and directly owned restaurants. Therefore, our sample only includes chains that franchise at least 5%, but no more than 95%, of their units in Florida.¹⁹ In Table 5, we implement specification (2), as discussed in Section 3.2. The dependent variable is the number of critical violations, and we are interested in the interaction *PostPE* * *DirectOwn*, where *DirectOwn* equals one if the store is owned by the chain. This specification estimates the differential treatment effect on directly owned versus franchised units. We include ZIP-code-by-year fixed effects in all specifications.

In Column (1), the interaction term is negative and significant. The coefficient on *PostPE* * *DirectOwn* is -0.317 , whereas the coefficient on *PostPE* alone is still negative but insignificant. Thus, the reduction in critical violations is concentrated in directly owned stores. Note that *DirectOwn* is insignificant, consistent with the ownership types being observably similar prior to buyout. In Column (2), we include the number of employees and seats: the results are similar, suggesting that improvements are not driven by changes to these physical features. Importantly, these improvements are not driven by unobservables such as brand or food genre, as these are similar for all restaurants within a chain, and these unobservables are absorbed by the store fixed effects. Overall, these results suggest that within the organization, improvements in operational practices are concentrated in stores in which PE has greater control and influence, consistent with hypothesis 1. Moreover, improved hygiene and the reduced likelihood of food poisoning generate positive externalities for all stores in the chain, and thus, consistent with hypothesis 2, this is a dimension along which we expect a gap between the two store ownership types.

¹⁸ In the Appendix, Table A5, we explore how the PE effect on critical violations varies across different private equity transaction types. As in Table 4 in the paper, we also include store fixed effects and ZIP-code-by-year fixed effects. Across all types of private equity buyouts, we find a statistically significant decline in the number of critical violations following the buyout. The largest decline occurs in private-to-private deals. However, estimates are not significantly different from those for secondary buyouts or divisional buyouts. The largest improvement appears to be in private-to-private deals. This is consistent with the findings of Bloom, Sadun, and Van Reenen (2009) that suggest that family firms seem to have the worst management practices and thus the most substantial room for improvement.

¹⁹ Results are similar if we use a 10% top/bottom cutoff.

Table 5
Inspection results by store ownership type with covariate interactions

	Critical violations	Critical violations	Critical violations	Critical violations	Critical violations	Critical violations	Critical violations
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
PostPE * DirectOwn	−0.317** (0.141)	−0.319** (0.141)	−0.332** (0.137)	−0.314** (0.141)	−0.334** (0.142)	−0.351*** (0.121)	−0.354*** (0.121)
PostPE	−0.16 (0.266)	−0.159 (0.266)	0.417 (0.417)	−0.0427 (0.310)	−0.146 (0.265)	0.605 (1.160)	0.374 (1.159)
DirectOwn	0.11 (0.122)	0.111 (0.122)	0.108 (0.122)	0.113 (0.122)	0.154 (0.122)	0.0895 (0.130)	0.13 (0.130)
Log(seats)		0.150* (0.086)	0.167* (0.085)	0.150* (0.086)	0.169* (0.087)	0.186* (0.101)	0.195* (0.101)
Log(employees)		−0.023 (0.025)	−0.014 (0.027)	−0.0234 (0.024)	−0.026 (0.026)	−0.019 (0.025)	−0.021 (0.031)
Interaction with PostPE	−	−	Seats & employment	County income	Check size	Cuisine type	All
Store fixed effects	X	X	X	X	X	X	X
ZIP × Year fixed effects	X	X	X	X	X	X	X
Observations	179,286	179,286	179,286	179,286	175,163	175,460	172,679
R ²	0.519	0.519	0.519	0.519	0.519	0.52	0.52

This table presents results from OLS regressions of critical violations on private equity ownership and store characteristics. An observation is an inspection on a specific date at a specific restaurant address. The independent variable DirectOwn is a dummy variable that equals one if the restaurant is owned and operated by its brand's parent company in a given year. DirectOwn equals zero if the restaurant is run by an independent franchisee. In Columns (3) to (7), the regressions include PostPE interacted with various characteristics of each individual store. Those characteristics—Seats & Employment, County Income, Check Size, and Cuisine Type—are defined and given in Table 2. Standard errors are in parentheses and clustered by restaurant chain. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Figure 2 displays the evolution of operational practices of franchises and directly owned stores around private equity takeover. The light bars plot the coefficients from a regression in which critical violations are regressed on private equity entry event-year dummies for directly owned stores only.²⁰ The critical violation trend is flat in the three years before PE entry, mitigating concerns that private equity targets chains because of an improving trend in its directly owned stores. The decline in critical violations then steadily occurs over the subsequent four years, becoming statistically significant from year two onward. In contrast, the dark bars plot the evolution of critical violations for franchises around the PE buyout. As with directly owned stores, no trends exist in the years leading to the buyout. Franchises, however, do not improve their operational practices in the years following the buyout, as none of the event year coefficients are statistically different from zero.

In the remainder of Table 5 we further investigate the robustness of the results documented in Columns (1) and (2) by exploring whether the postbuyout gap between directly owned stores and franchises can be attributed to alternative store characteristics. In particular, we repeat the specification outlined in Column (2) and add interaction terms of the *PostPE* variable

²⁰ This regression is identical to the one reported in Column (1) of Table A4, but the sample is restricted to directly owned stores. The regression includes store fixed effects, ZIP-code-by-year fixed effects, log number of seats, and log number of employees. The regression results are in Column (3) of Table A4 in the Appendix. The average number of critical violations was added to coefficients in the graph to illustrate the relative size of the coefficients.

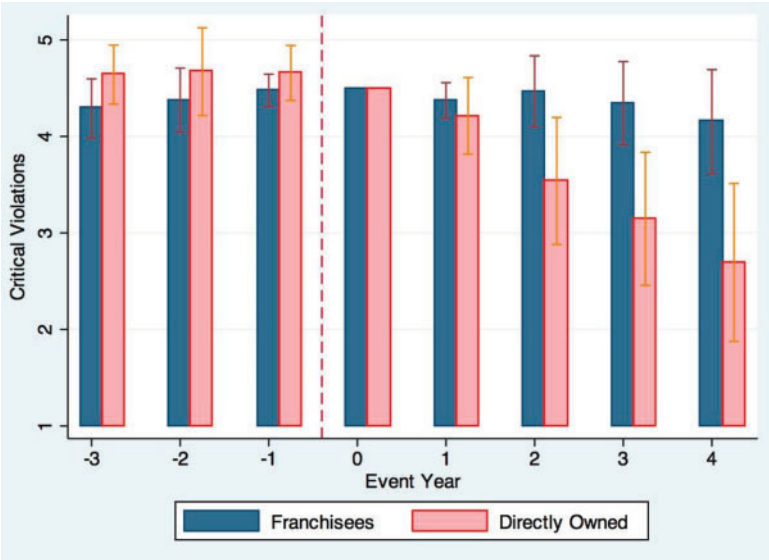


Figure 2
Franchisees and directly owned restaurants around private equity deal date

This figure plots the coefficients and 95% confidence interval bands of regressions of critical violations by directly owned and franchised stores on event year dummy variables around the date private equity acquires a restaurant. Additional control variables are restaurant fixed effects, ZIP-code-by-year fixed effects, number of employees, and number of seats. Standard errors are clustered at the level of the chain. Event-year zero is the omitted variable, corresponding to inspections that occur from 1 to 365 days after the deal close date. The average number of critical violations was added to coefficients in the graph.

with various characteristics such as size (employees and number of seats), county income, price category, and cuisine category. We add each of these interactions separately in Columns (3) to (6) and jointly in Column (7). The magnitude of the interaction variable of *PostPE* and directly owned stores remains unchanged regardless of the additional controls we provide. This suggests that the effect of PE on directly owned stores is almost orthogonal to other store-level characteristics. In fact, the results become more statistically significant, as the estimation is more precise.

4.3 Robustness checks

In this section, we discuss alternative explanations to our findings. Our first concern is that the reduction in critical violations and improvement in operational practices is not unique to private equity investors, but rather is mechanically associated with any change in ownership. Managers might pay particular attention to impress the new owner. Using our comprehensive annual ownership data from Florida restaurant licensing agreements, we flag an ownership change when the owner name changes, while the restaurant name does not. As reported in Table A6 in the Appendix, we do not find a decline in critical violations following restaurant ownership changes in general. However,

when ownership changes are driven by private equity buyouts, we find a large and statistically significant decline in critical violations after the buyout, ruling out a simple ownership change story.

Another concern is store attrition. Perhaps directly owned stores are more likely to be closed if performing poorly, leading to a survivorship bias and thus an alternative explanation of the results. In Table A7 in the Appendix we explore such a hypothesis. First, we find that, in general, stores are not more likely to close following the PE buyout. Moreover, in contrast to the concern, we do not find that directly owned stores are more likely to be closed relative to franchises following the PE buyout. In Table A8 in the Appendix we also study whether PE firms are more sensitive to critical violations in cases of store closures. We find that this is not the case, as PE-backed chains are not responding differently to lagged critical violations relative to other chains. Similarly, we do not find that within a chain, the response to lagged critical violations is different between franchises and directly owned stores.

Alternatively, one may be concerned that the ownership mix changes following the PE buyout. Evidence suggests that chains have an optimal mix of franchises and directly owned stores since their portfolio mix is stable and persistent over time (Lafontaine and Shaw 2005; Lafontaine and Kaufmann 1994). In Table A9 in the Appendix, we do not find that PE-backed chains deviate from this ownership mix, as they are not more likely to convert directly owned stores to franchises or franchises to directly owned stores. Moreover, in Table A10 in the Appendix we find that PE-backed chains are not more likely to open new directly owned stores or new franchises after the buyout, relative to other chains. Overall, we find that the composition of store ownership within a chain does not systematically change following a PE buyout relative to other restaurant chains.

Another potential explanation may be that private equity firms are better at teaching restaurant managers how to deal with inspectors, perhaps through persuasion or even manipulation and bribery. In such cases, reductions in critical violations may not represent real outcomes. We believe that this is unlikely since health inspection results in Florida are not salient to customers. Like in Los Angeles or New York, for example, in Florida, grade cards are not highly visible at restaurants. Results are only available online via the state of Florida Web site, and anecdotal evidence suggests customers rarely look up health scores before going out to eat. This means that inspection scores cannot immediately affect restaurant business. Moreover, fines due to health violations are marginal: in 2012, approximately 2% of inspections resulted in fines, with a median fine of \$400. Therefore, it seems that the primary benefit of reducing violations is the *actual* improvement in store operations, and only *appearing* to improve by exerting risky effort or money is of little benefit.

As discussed in Section 3.3, a restaurant chain does not franchise or directly run stores randomly. The literature provides several explanations for this within-chain decision. In particular, location seems to play an important role. This may

be a concern if locations in which direct ownership dominates are experiencing unobserved shocks correlated with private equity buyouts. We consider the two main theories discussed in the literature. First, restaurants farther from headquarters might be more difficult to monitor and hence given to a franchisee who is the residual claimant. This concern is mitigated by the inclusion of ZIP-code-by-year controls. Second, stores located near highway exits might be more frequently directly owned, as customer turnover is likely to be greater in these locations and franchisees do not internalize externalities. While ZIP-code-by-year fixed effects may alleviate some of this concern, we attempt to be more precise. We use ArcGIS software to identify stores located within 0.5 or 0.25 miles of a highway exit and construct a “near highway” dummy variable. As reported in Table A11 in the Appendix, the main results are unchanged when we interact “near highway” with the PostPE treatment effect, suggesting that this cannot explain the findings.

Perhaps franchises are better maintained than directly owned stores and the buyout occurs because of anticipation that directly owned stores will converge to these higher standards, explaining the more pronounced directly owned decline in critical violations. This argument is inconsistent, however, with the finding of no pre-existing trends for either franchises or directly owned stores leading to the buyout. In Table 1, panel F, we do not find that franchisees are better maintained, as the number of critical violations, on average, is slightly lower in directly owned stores.

What if the results are driven by adverse changes in the contractual agreements of headquarters with the franchisees? To explore whether this is the case, we manually collect information on franchise contracts from *Entrepreneur* magazine’s “Franchise 500 Ranking” for the years 2004 to 2015. The information provided includes royalty rates, franchise fees, franchisee startup costs, and whether franchisors provide financing. In Table A12 in the Appendix, we explore whether any of these contractual dimensions change around PE transactions, and we find they do not. This is consistent with Lafontaine and Shaw (1999), who find that contract terms are persistent over time within firms and conclude that variation in contract terms is mostly determined by differences across firms.

5. Underlying Channels

Thus far, we find that private equity firms improve operational practices in restaurant chains they acquire, as restaurants become cleaner, safer, and better maintained. The effect is concentrated in directly owned stores, which is consistent with the first and second hypotheses. In this section we further explore the mechanism through which PE firms affect restaurant chains. First, we explore whether PE firms hire additional workers to generate these operational improvements. We find, in fact, that store-level employment declines throughout the entire chain. This is also consistent with the second

Table 6
Restaurant employment under private equity ownership

	Employees	Employees	Employees	Employees
	(1)	(2)	(3)	(4)
PostPE	−0.028** (0.013)	−0.024** (0.012)	−0.024** (0.012)	−0.018* (0.011)
PostPE * DirectOwn			0.005 (0.011)	−0.005 (0.009)
DirectOwn			0.008 (0.010)	0.018* (0.011)
Log(seats)	0.026*** (0.008)	0.033*** (0.008)	0.014* (0.008)	0.021** (0.008)
Store fixed effects	X	X	X	X
Year fixed effects	X		X	
ZIP × Year fixed effects		X		X
Observations	246,387	246,387	74,345	74,345

This table presents results from OLS regressions of restaurant employment on private equity ownership. An observation is a store-year. The dependent variable *Employees* is the log of the average number of full-time equivalent employees at a store in a given year. The remaining variables are as defined in Table 4. Standard errors are in parentheses and clustered by restaurant chain. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

hypothesis that for cost reduction initiatives, franchisees are more likely to adopt PE initiatives. Second, we also find support for the third hypothesis, as PE-backed chains with relevant expertise and experience show greater operational improvements.

5.1 Workforce

An increase in employment might be a mechanism through which PE firms affect operational practices. PE impact on employment, generally, is controversial. The popular press often chides them for eliminating jobs for debt service and short-term profits, whereas Davis et al. (2014) find that private equity transactions result in only a modest net impact on employment.

We explore the effect on this stakeholder in Table 6. The dependent variable is the log of the number of employees at a store. In Column (1), we include year fixed effects and explore variation within a store by adding individual store fixed effects. The coefficient on *PostPE* is negative and significant, suggesting that PE firms appear to operate existing restaurants with fewer employees than before. The magnitude, however, is fairly modest. The coefficient is −0.028, suggesting a 2.8% decline in a store’s workforce. This decline is in line with the findings in Davis et al. (2014) and inconsistent with additional hiring to support changes to operational procedures. To control for the possibility that PE targets are located in areas that, perhaps due to varying economic conditions, have employment patterns different from other restaurants, we include ZIP-code-by-year fixed effects in Column (2). PE restaurants still see a decline in workers when adjusting for geographic variation and local shocks. In Columns (3) and (4) of Table 6, we include the *PostPE* * *DirectOwn* interaction to see if the employment effect is stronger in directly controlled branches. We restrict the sample to chains that employ franchising for at least 5% of its units and

no more than 95% of its units in Florida, as in Section 4.2. The interaction is essentially zero, meaning both company-owned and franchised outlets see a similar decline in headcount.

This finding is consistent with hypothesis 2: reducing store-level employee headcount can generate improved profitability for a given store but is unlikely to generate positive externalities for the chain. Hence, franchisee and parent interests are congruent, and we expect a smaller (or no) gap in implementation between the two ownership types since franchisees will be willing to adopt such initiatives.

5.2 Private equity partner experience

Private equity owners may use expertise and knowledge to improve firm operations. In Table A3 in the Appendix we find that the operational improvements are strongest in areas related to food handling. These are arguably “soft” skills (e.g., avoiding cross-contamination, maintaining proper temperatures) not easily sharpened simply by capital reallocation. Rather, knowledge of better techniques and more effective training, enforcing, and monitoring of store employees is required. Consistent with this story, Bloom, Sadun, and Van Reenen (2009) find that PE-backed firms are associated with better management practices. We use a private equity partner’s past experience as a proxy for the relevant human capital they bring to restaurant management. This follows from Acharya et al. (2013), who find that partner background relates to the performance of the portfolio company: general partners with an operational background are better at internal, organic value creation, while those with a finance background generate greater outperformance through M&A. We test hypothesis 3 that partner operational experience, or even more specifically, restaurant experience, is associated with better health inspection outcomes.

For each deal in our sample, we manually collect the names of all PE firm partners mentioned in the acquisition press release. We identify at least one responsible partner and, occasionally, two or three, for 80% of the sample deals. Deals for which no partners are identified are removed from this analysis. We use PE firm Web sites and other Internet searches to determine the work history of each partner. The variable *OperatingExp* equals one for restaurants acquired by PE firms with at least one partner who worked at a consulting firm or in industry prior to joining their current PE firm. The variable *RestaurantExp* equals one when a partner has restaurant industry experience in particular. If *RestaurantExp* equals one, *OperatingExp* necessarily also equals one.

In Table 7 critical violations are regressed on *PostPE* and an interaction with the past work experience variables.²¹ *PostPE* is negative and significant

²¹ We do not explore the interaction of partner background with direct ownership since no clear prediction or interpretation for the interaction term exists. Greater experience could improve the overall level of operational practices in the chain, leaving the gap between franchises and directly owned stores unchanged. Alternatively, expertise could increase the gap if improvements accrue disproportionately to stores in which PE firms have

Table 7
Private equity partner backgrounds and inspection outcomes

	Critical violations	Critical violations	Critical violations	Critical violations
	(1)	(2)	(3)	(4)
PostPE	−0.541*** (0.157)	−0.540*** (0.157)	−0.430*** (0.142)	−0.429*** (0.142)
PostPE * OperatingExp	−0.391 (0.527)	−0.390 (0.527)		
PostPE * RestaurantExp			−1.142*** (0.418)	−1.142*** (0.428)
Log(seats)		0.151*** (0.055)		0.151*** (0.055)
Log(employees)		−0.025 (0.020)		−0.026 (0.020)
Store fixed effects	X	X	X	X
ZIP × Year fixed effects	X	X	X	X
Observations	529,673	529,673	529,673	529,673

This table presents results from OLS regressions of critical violations on private equity ownership and private equity firm partner characteristics. An observation is an inspection on a specific date at a specific restaurant address. The independent variable *OperatingExp* is a dummy variable that equals one if the lead partner from the private equity firm acquiring the restaurant had either consulting or industry experience prior to joining the PE firm. *RestaurantExp* equals one if the lead partner had restaurant industry experience prior to joining the PE firm. The remaining variables are as defined in Table 4. Standard errors are in parentheses and clustered by restaurant chain. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

in Column (2) at -0.54 ; thus, the main results hold without the interaction with partner background. The coefficient on *PostPE*OperatingExp*, though insignificant, is negative, with a slightly lower magnitude at -0.39 . Thus, directionally, PE deals led by operational partners result in greater improvement in restaurant practices. In Column (4), the coefficient on *PostPE*RestaurantExp* is -1.14 and significant at the 1% level; deals led by partners who have previously run restaurants exhibit dramatic target improvements. These results suggest a channel for operational improvements is application of know-how and PE firm expertise.

6. Conclusion

We study the operational consequences of private equity buyouts in the restaurant industry. We find that restaurants improve operational practices following a PE buyout and commit fewer health violations. As a consequence, restaurants are safer, cleaner, and better maintained. Practices improve more in stores in which PE firms have direct ownership and hence more influence and control. Franchises, which are otherwise similar, do not see the same improvement, suggesting that PE firms cause these changes and are not merely selecting restaurants on an upward trajectory.

more control. Lastly, more experienced partners may know how to handle franchisees better, shrinking the gap between franchises and directly owned stores.

This paper provides new evidence on an operational engineering channel through which private equity firms add value. The nature of health violations suggests that PE firms improve management practices through training, monitoring, and alignment of worker incentives and not only by alleviating financial constraints and reallocating capital. In support of PE firm managerial skill, we find that they improve stores more when led by partners with prior restaurant experience.

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